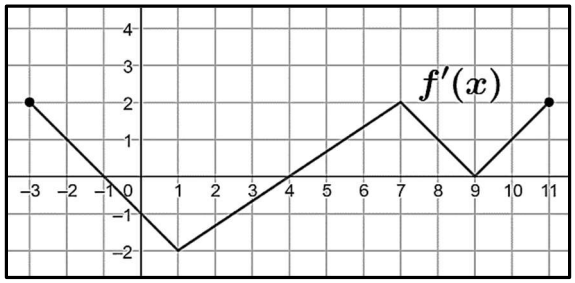
Worksheet 5-4 A Relative Extrema #1
AP Calculus Name_____

Directions: Use the following information to answer the given questions. Show all work.



1. The graph of f', the derivative of f, is shown above for $[-3, 11]$ and consists of four line segments. Use the graph of f' to answer the following.

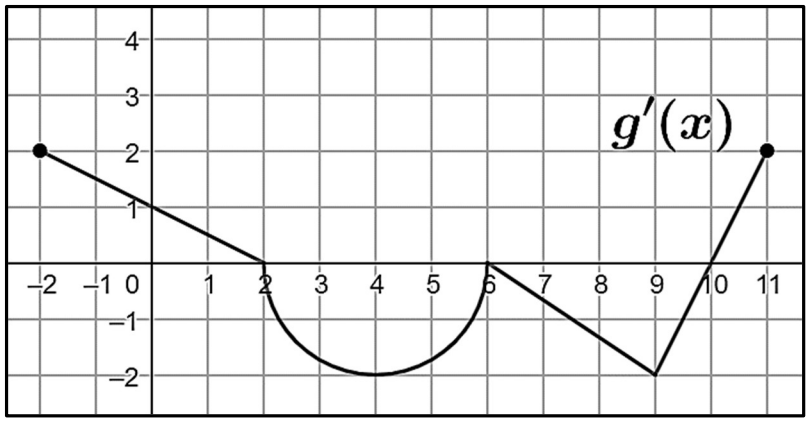
A) Find all critical values of $f(x)$ in the open interval $(-3, 11)$.

B) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = 7$? Give a reason for your answer.

C) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = -1$? Give a reason for your answer.

D) Does $f(x)$ have a relative minimum, relative maximum, or neither at $x = 9$? Give a reason for your answer.

E) On what interval(s) in $[-3, 11]$ is $f(x)$ increasing? Give a reason for your answer.



2. The graph of g', the derivative of g, is shown above for $[-2, 11]$ and consists of three line segments and a semicircle. Use the graph of g' to answer the following.

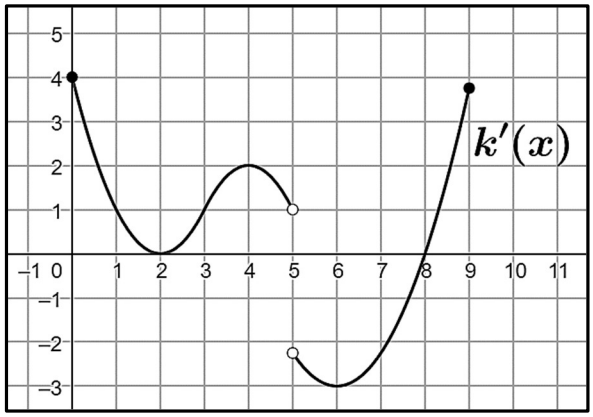
A) Find all critical values of $g(x)$ in the open interval $(-2, 11)$.

B) At which value(s) of x does $g(x)$ have a local minimum? Explain your reasoning.

C) On what interval(s) in $[-2, 11]$ is $g(x)$ increasing? Give a reason for your answer

D) Does $g(x)$ have a local minimum, local maximum, or neither at $x = 4$? Give a reason for your answer.

E) Let $h(x) = xg(x)$. If $h(x)$ has a relative maximum at $x = 4$, find $h(4)$. Show the work that leads to your answer.



3. The function k is continuous on the interval $[0, 9]$. The graph of $k'(x)$ is given above and has horizontal tangents at $x = 2, x = 4,$ and $x = 6$.

A) On which interval(s) is $k(x)$ increasing? Give a reason for your answer.

B) At which value(s) of x does $k(x)$ have a relative minimum? Explain your reasoning.

C) At which value(s) of x does $k(x)$ have a local maximum? Explain your reasoning.

D) Let $m(x) = k(x) - \sin(x - 2)$. Find $m'(2)$. Does $m(x)$ have a local minimum, local maximum, or neither at $x = 4$? Give a reason for your answer.

Directions: Given the following information, find and classify the x-values of any relative extrema of the given function. Justify your answers.

4. $f(x) = -x^3 + 6x^2 - 9x + 7$

5. $f'(x) = 3x(x - 4)(x + 1)$

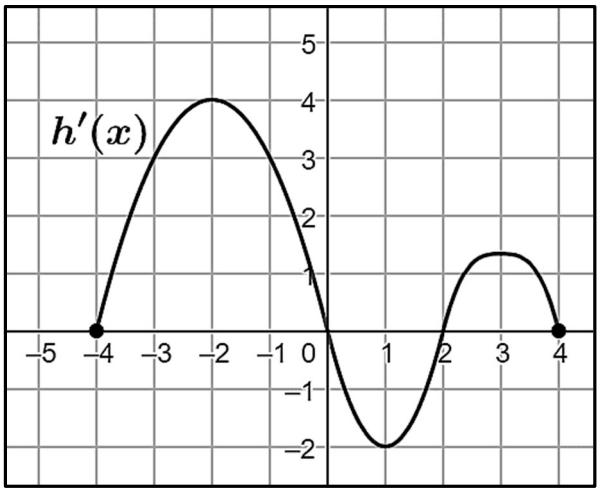
6. $f(x) = x^2 e^x$

7. $f'(x) = 2\sin(x) + 1$ on $[0, 2\pi)$

8. $g'(x) = \dfrac{x-3}{e^x}$

9. $g(x) = \dfrac{x-3}{x^2}$

10. The function h is twice differentiable on the interval $[-4, 4]$. The graph of $h'(x)$ is given at the right and has horizontal tangents at $x = -2, x = 1$, and $x = 3$. It is known that $h(-1) = 5$.



A) On which interval(s) is $h(x)$ decreasing? Give a reason for your answer.

B) Find all critical values for the function $h(x)$. Classify each critical value as a local minimum, local maximum, or neither. Explain your reasoning.

C) Let $k(x) = h(x) - 3x$. At which x value(s) does $k(x)$ have a relative maximum? Explain your reasoning.

D) Find the value of $\lim_{x \to -1} \dfrac{h'(x^2) - 2x}{h(x) - 5}$ or state that it does not exist. Justify your answer.

x	$-4 \le x < -1$	-1	$-1 < x < 3$	3	$3 < x < 8$	8	$x > 8$
$f(x)$	negative	-8	negative	0	positive	2	positive
$f'(x)$	negative	0	positive	0	positive	undefined	negative

11. The function f is continuous on the interval $[-4, \infty)$. The table above gives selected information about f and its derivative f'. It is known that $f(x) < 3$ for all values of x in its domain.

A) On what interval(s) is the function $f(x)$ increasing? Give a reason for your answer.

B) Find any critical values for the function $f(x)$. For each critical value, determine if it is the location of a relative minimum, a relative maximum, or neither for $f(x)$. Justify your answers.

C) Let $C'(x) = (4 - x)f'(x)$ be the derivative of the function $C(x)$. Does C have a local minimum, local maximum, or neither at $x = 4$. Explain your reasoning.

E) Let $n(x) = f(x^2)$. Does $n(x)$ have a local minimum, local maximum, or neither at $x = -1$. Justify your answer.